

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 915+100

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – VUP

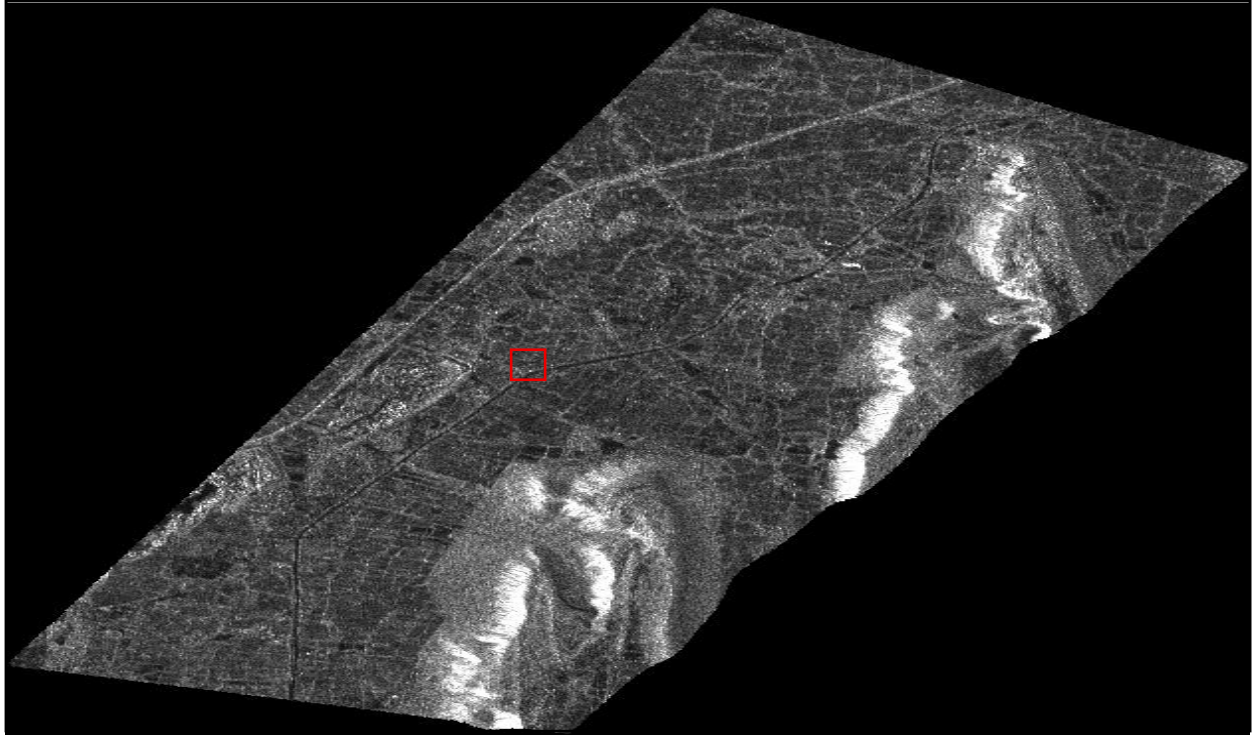
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

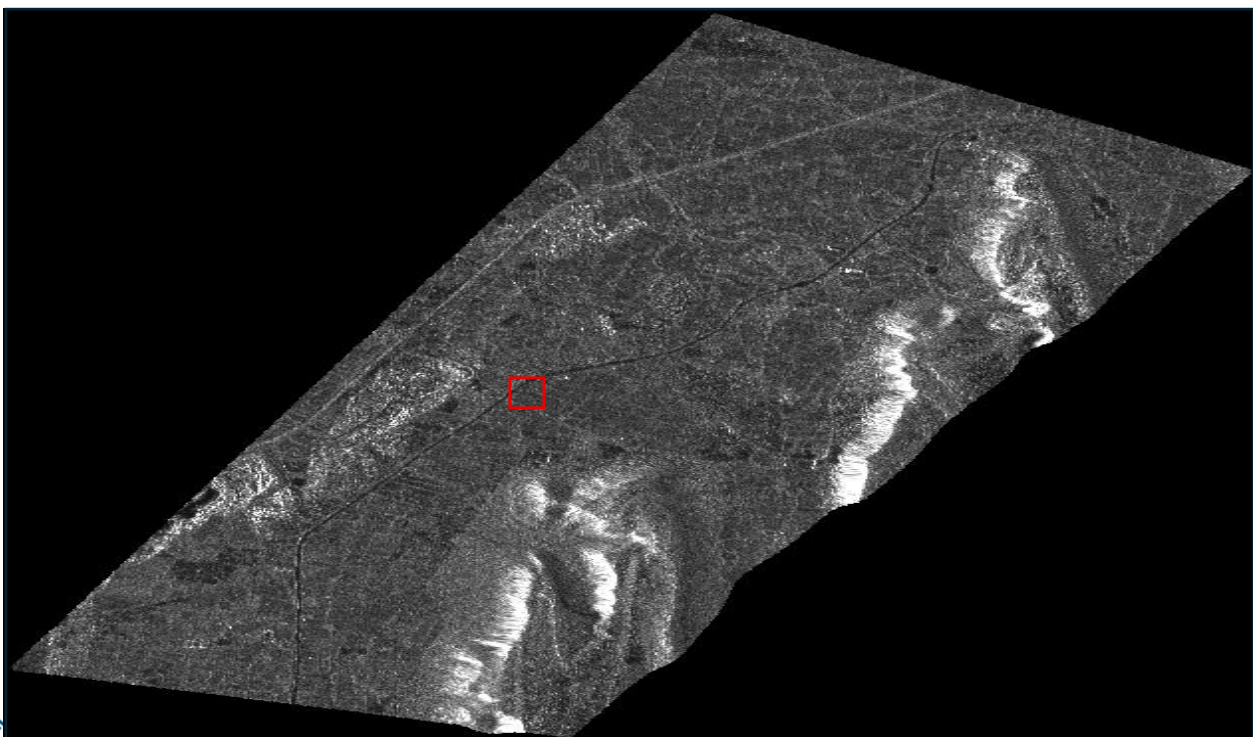
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

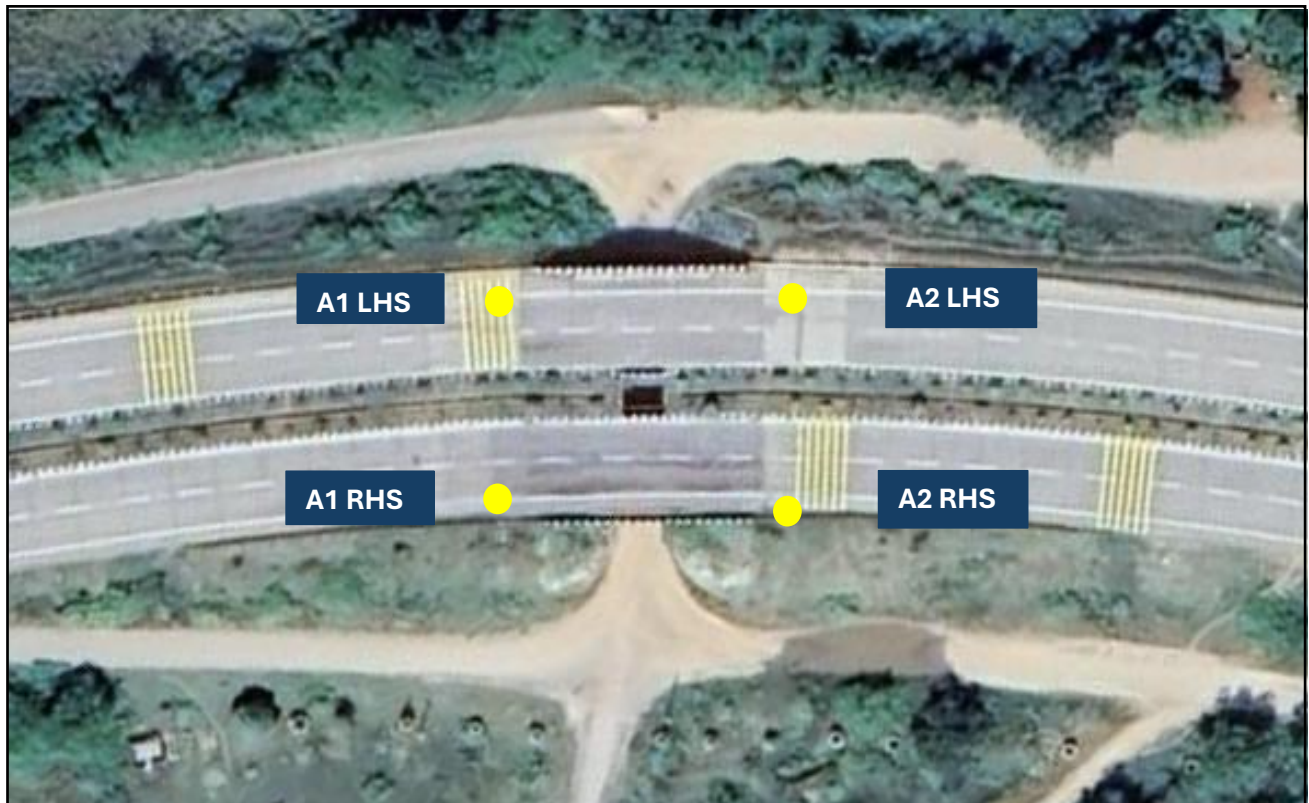


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

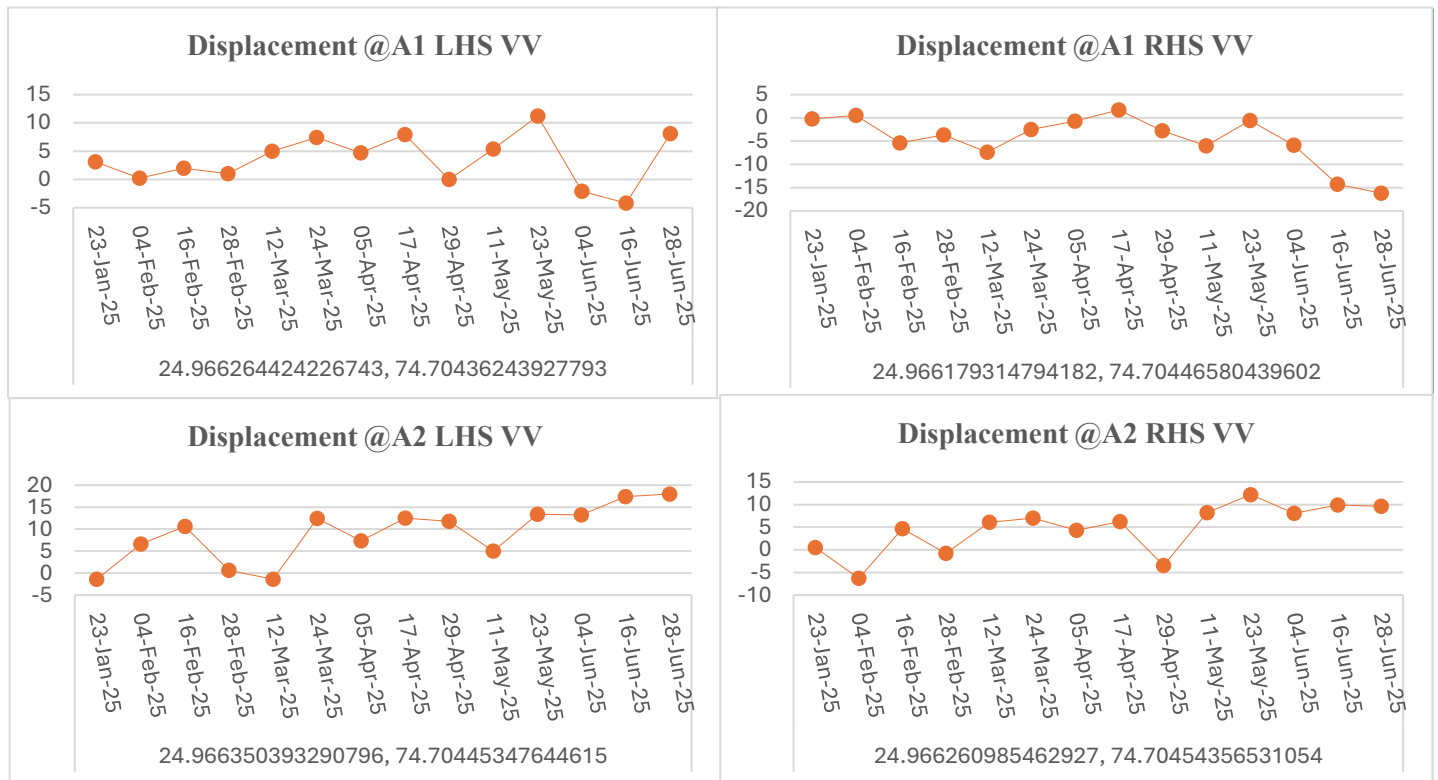
VUP – CH-915+100

Length – 20.47M

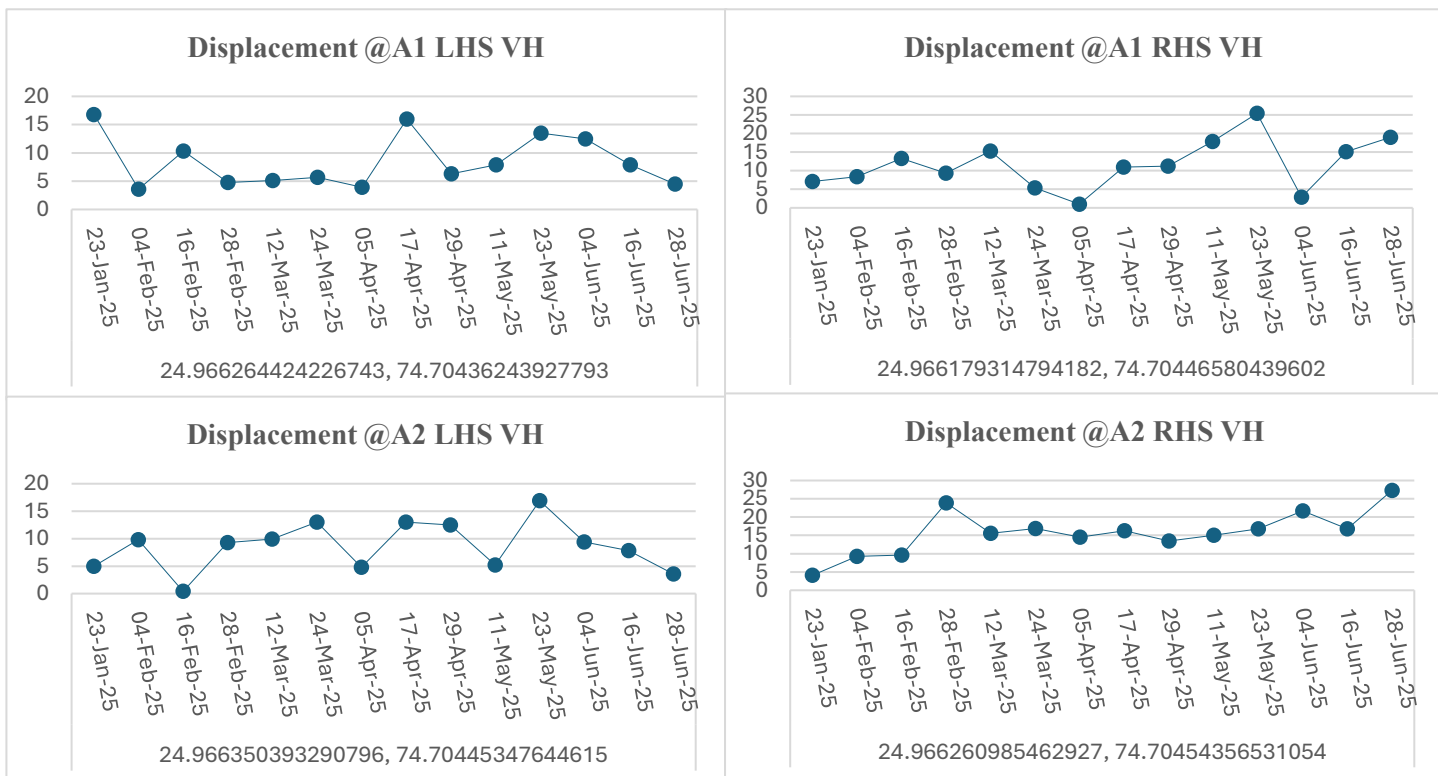


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



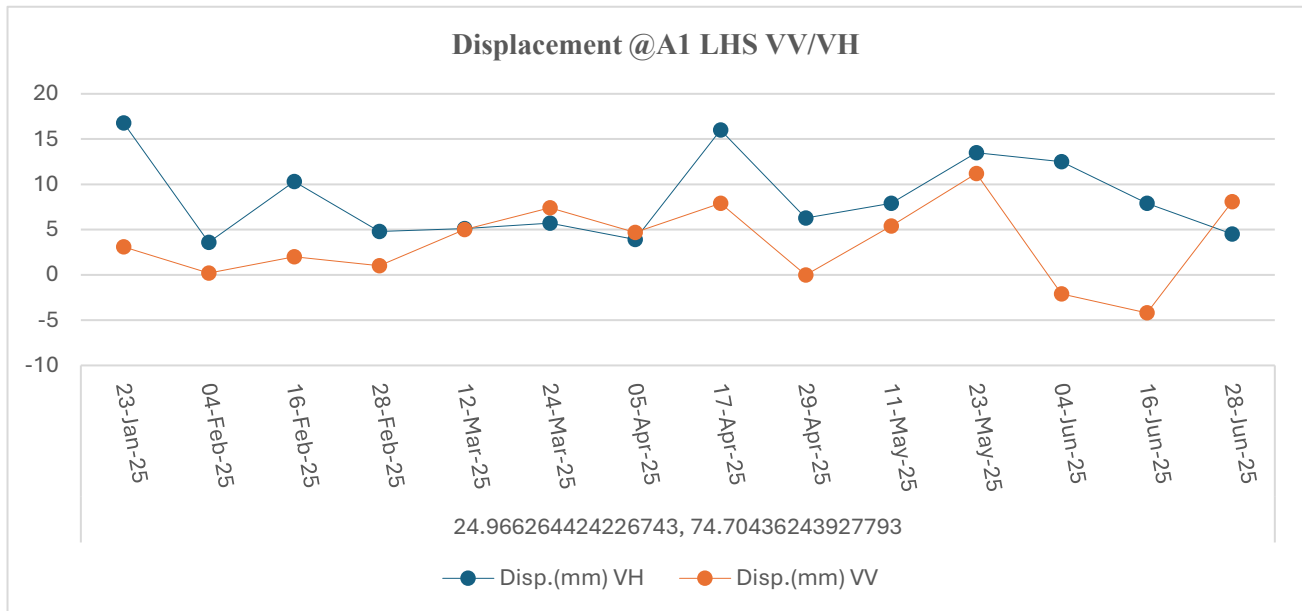
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.966264424226743, 74.70436243927793)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** exhibits moderate variability with alternating uplift and settlement phases throughout the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate-to-high variability with predominantly positive values, ranging roughly between +3 mm and +16 mm

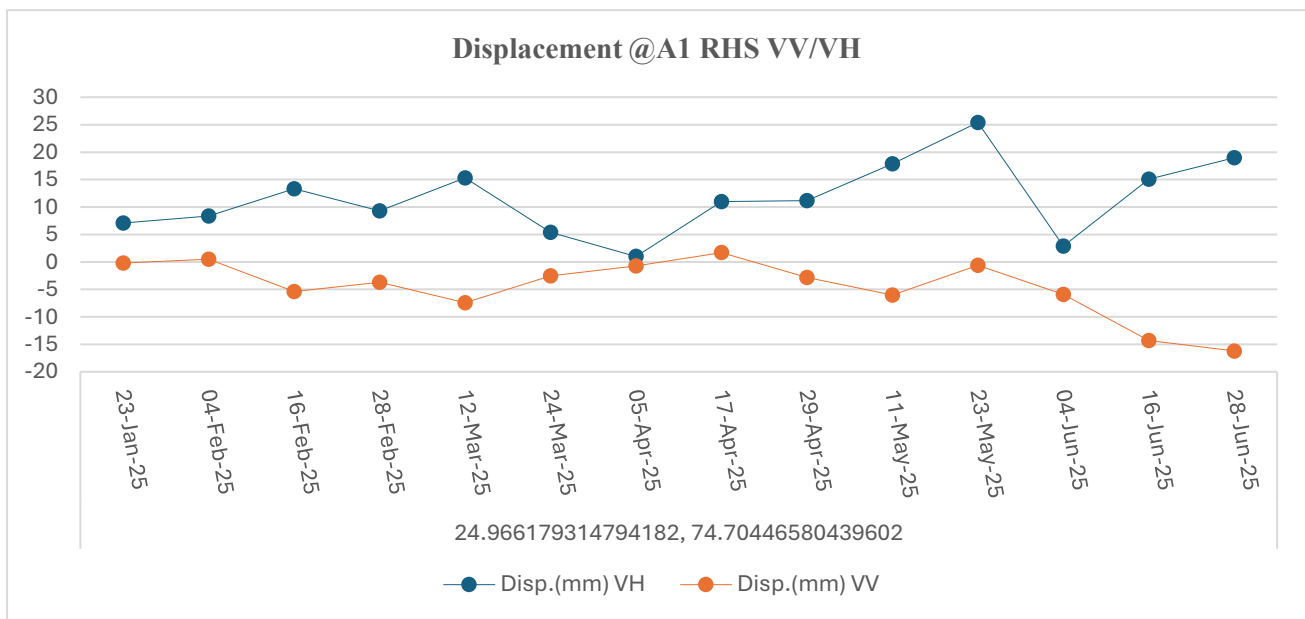
Critical Events & Anomalies:

- **January 2025:** VH begins near +16 mm, while VV is around +3 mm, indicating a strong initial lateral displacement with mild vertical uplift.
- **May 2025:** VH rises to +8 mm, while VV increases to +5 mm, marking renewed uplift activity.
- **June 2025:** VH reduces further to +5 mm, while VV rebounds to +8 mm, indicating vertical recovery by the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.966179314794182, 74.70446580439602)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows high variability with a strong positive dominance, ranging roughly between +4 mm and +26 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits predominantly negative behaviour with moderate variability, indicating overall vertical settlement at the A1 RHS location throughout most of the monitoring Period.

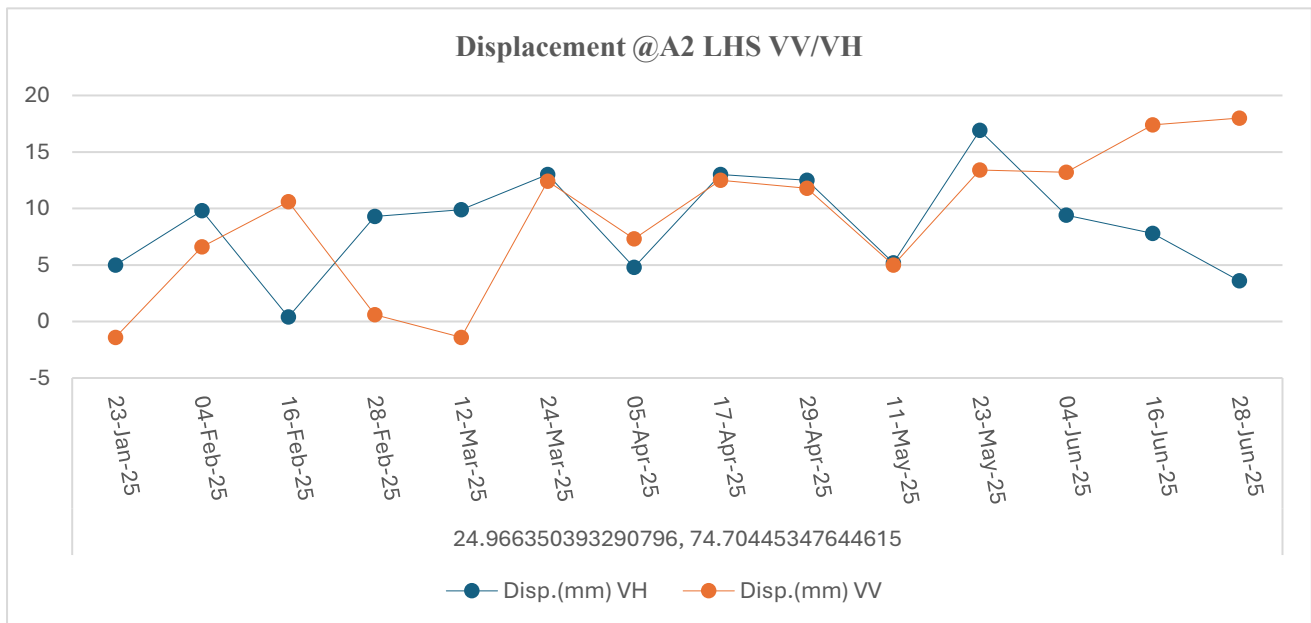
Critical Events & Anomalies:

- **January 2025:** VH begins near +7 mm, while VV is close to 0 mm, indicating initial lateral displacement with negligible vertical movement.
- **April 2025:** VH remains near +11 mm, while VV drops again to -3 mm, suggesting loss of vertical recovery under sustained lateral movement.
- **June 2025:** VH recovers to +15 mm, while VV deteriorates further to -13 mm, indicating persistent vertical settlement despite lateral recovery.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.966350393290796, 74.70445347644615)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with alternating uplift and reduction phases, ranging roughly between 0 mm and +17 mm.
- **Vertical Displacement (VV)** Vertical displacement shows moderate-to-high variability with a clear positive dominance, indicating overall vertical uplift at the A2 LHS location during the monitoring period.

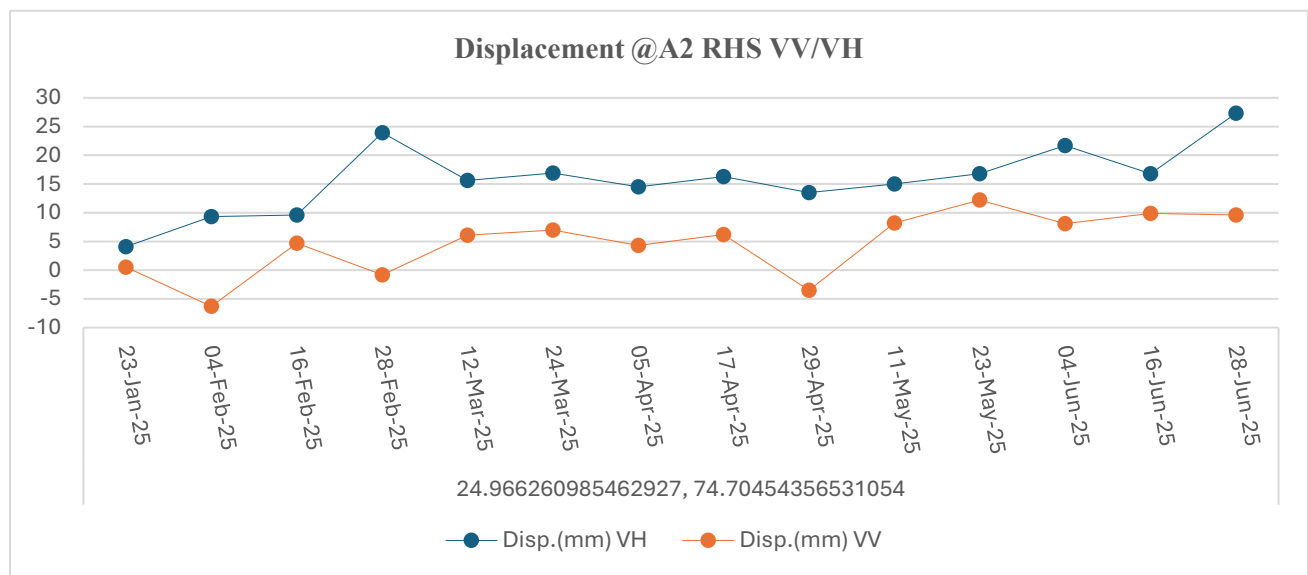
Critical Events & Anomalies:

- **January 2025:** VH begins near +5 mm, while VV is slightly negative, indicating initial lateral displacement with minor vertical settlement.
- **June 2025:** VH gradually decreases to +4 mm, while VV continues to increase to +18 mm, highlighting strong vertical uplift dominance toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.966260985462927, 74.70454356531054)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits moderate variability with a predominantly positive trend, indicating overall vertical uplift at the A2 RHS location during most of the monitoring period.
- **Horizontal (VH) displacement** Horizontal displacement shows high variability with a strong positive dominance, ranging roughly between +5 mm and +27 mm.

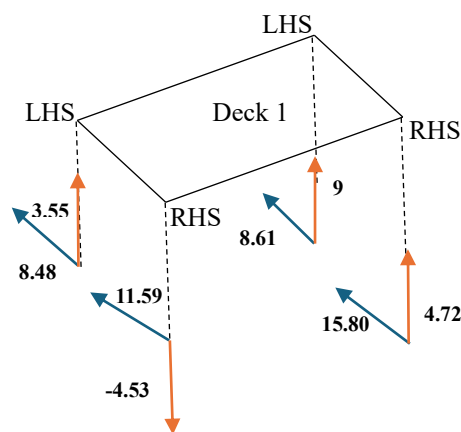
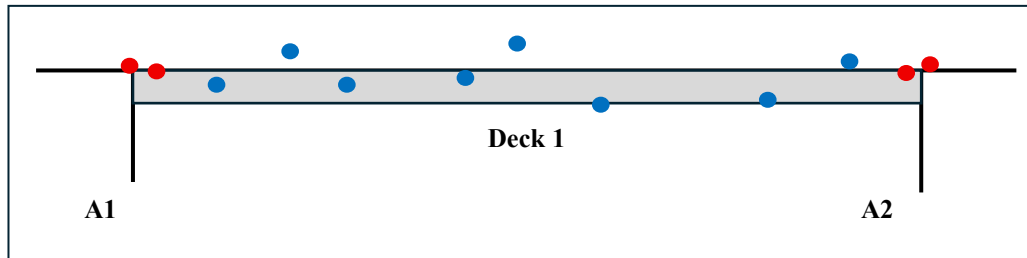
Critical Events & Anomalies:

- **January:** VH starts near +5 mm, while VV is close to +1 mm, indicating initial mild uplift in both horizontal and vertical components.
- **May 2025:** VH increases to +15 mm, while VV rebounds strongly to +9 mm, indicating recovery from the Late-April disturbance.
- **June 2025:** VH reaches +22 mm, while VV reduces slightly to +8 mm, suggesting lateral acceleration with mild vertical reduction.

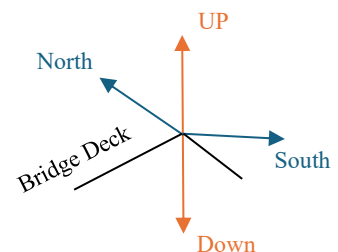
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The coexistence of uplift at both supports and settlement at the central lower region confirms vertical rotation combined with localized bending, rather than uniform vertical translation.

Horizontal Displacement Behaviour:

- Horizontal displacement shows strong directional variability across the deck, reflecting a torsion-dominated deformation pattern. The maximum horizontal displacement occurs at the RHS lower region, reaching approximately +15.80 mm, indicating dominant lateral movement on the RHS.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-controlled deformation response, characterized by simultaneous vertical uplift, localized settlement, and significant asymmetric horizontal displacement. Overall, the observed displacement pattern is structurally significant, warranting continued monitoring and correlation with loading conditions, bearing behaviour, and boundary constraints.